

Lab 04 –

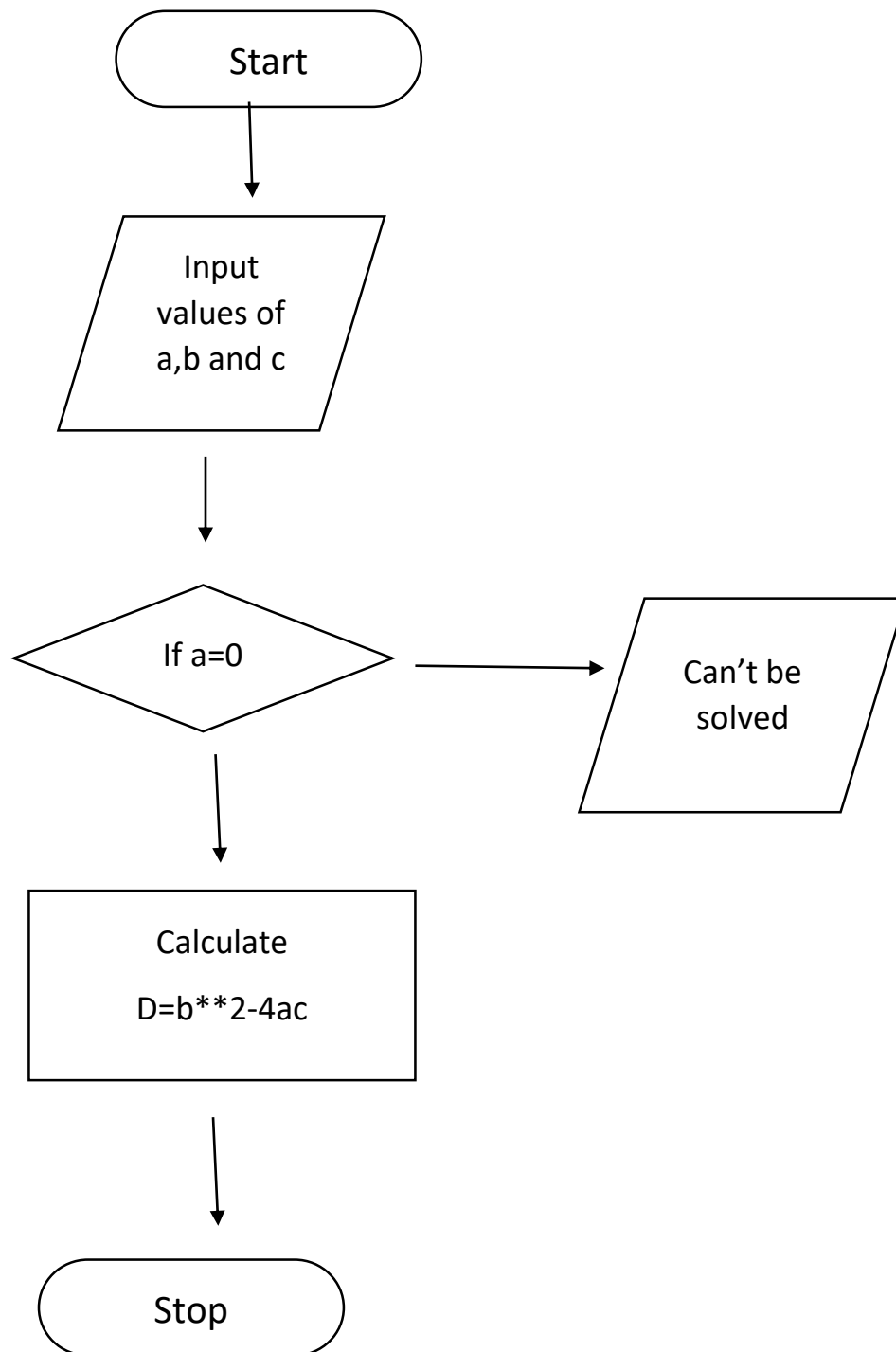
Conditional Statements and Simple Loops

Programming Exercise:

1. Write a program which solves the quadratic equation. The user will enter the value of a, b and c. The program will then check the denominator that if denominator is zero or not. If its zero it can reply the equation cannot solve as there is a zero division else, it will execute the program and will generate two solutions.

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Flowchart:



Source Code:

```
1  import cmath # For complex square root calculation
2
3  # Loop until valid input is given
4  while True:
5      try:
6          a = float(input("Enter the coefficient a: "))
7          b = float(input("Enter the coefficient b: "))
8          c = float(input("Enter the coefficient c: "))
9
10         # Check if 'a' is zero
11         if a == 0:
12             print("The equation cannot be solved as 'a' cannot be zero.")
13             continue # Ask for input again
14
15         # Calculate the discriminant
16         discriminant = b ** 2 - 4 * a * c
17
18         # Check if the discriminant is negative
19         if discriminant < 0:
20             print("The equation has complex solutions.")
21             # Calculate complex solutions
22             solution1 = (-b + cmath.sqrt(discriminant)) / (2 * a)
23             solution2 = (-b - cmath.sqrt(discriminant)) / (2 * a)
24             print(f"The solutions are: x1 = {solution1}, x2 = {solution2}")
25         else:
26             # Calculate the two real solutions
27             solution1 = (-b + discriminant ** 0.5) / (2 * a)
28             solution2 = (-b - discriminant ** 0.5) / (2 * a)
29             print(f"The solutions are: x1 = {solution1}, x2 = {solution2}")
30
31         # Break the loop after successful calculation
32         break
33
34     except ValueError:
35         print("Invalid input! Please enter numeric values.")
36
```

Output:

```
C:\Users\hp\PycharmProjects\main\venv\Scripts\python.exe C:\Users\hp\PycharmProjects\main\main.py
Enter the coefficient a: 0
Enter the coefficient b: 4
Enter the coefficient c: 9
The equation cannot be solved as 'a' cannot be zero.
Enter the coefficient a: 5
Enter the coefficient b: 7
Enter the coefficient c: d
Invalid input! Please enter numeric values.
Enter the coefficient a: 4
Enter the coefficient b: 6
Enter the coefficient c: 7
The equation has complex solutions.
The solutions are: x1 = (-0.75+1.0897247358851685j), x2 = (-0.75-1.0897247358851685j)

Process finished with exit code 0
```

2. Calculate the arithmetic sequence of n numbers. The program will generate the nth term of the sequence, whereas the user will enter the first term and the common difference. The program will then ask either to continue or not, if the user will enter yes it will ask the next nth term to calculate.

Source Code:

```
1  # Program to calculate the nth term of an arithmetic sequence
2
3  while True:
4      # Get user inputs
5      try:
6          first_term = float(input("Enter the first term (a1): "))
7          common_difference = float(input("Enter the common difference (d): "))
8          n = int(input("Enter the term number (n) you want to calculate: "))
9
10         # Calculate the nth term using the formula
11         nth_term = first_term + (n - 1) * common_difference
12
13         # Display the result
14         print(f"The {n}th term of the arithmetic sequence is: {nth_term}")
15
16         # Ask if the user wants to continue
17         continue_prompt = input("Do you want to calculate another term? (yes/no): ").strip().lower()
18
19         if continue_prompt != 'yes':
20             print("Thank you for using the program!")
21             break # Exit the loop if the user does not want to continue
22
23     except ValueError:
24         print("Invalid input! Please enter numeric values.")
```

Output:

```
C:\Users\hp\PycharmProjects\main\venv\Scripts\python.exe C:\Users\hp\PycharmProjects\main\main2.py
Enter the first term (a1): 1
Enter the common difference (d): 2
Enter the term number (n) you want to calculate: 4
The 4th term of the arithmetic sequence is: 7.0
Do you want to calculate another term? (yes/no): Yes
Enter the first term (a1): 2
Enter the common difference (d): d
Invalid input! Please enter numeric values.
Enter the first term (a1): 2
Enter the common difference (d): 2
Enter the term number (n) you want to calculate: 5
The 5th term of the arithmetic sequence is: 10.0
Do you want to calculate another term? (yes/no): no
Thank you for using the program!

Process finished with exit code 0
```

3. Write a program which will check either the giving string is Palindrome or not. Palindrome is a string when we reverse the string it will generate the original string. Example CIVIC, MOM, 010, 1001, etc. So if you enter the word which is Palindrome it will say yes your string is Palindrome otherwise it will generate sorry message.

Source Code:

```
1  # Program to check if a string is a palindrome
2
3  while True:
4      # Get user input
5      original_string = input("Enter a string to check if it is a palindrome: ")
6
7      # Remove spaces and convert to lowercase for uniformity
8      cleaned_string = original_string.casefold()
9
10     # Reverse the cleaned string
11     reversed_string = cleaned_string[::-1]
12
13     # Check if the original string is the same as the reversed string
14     if cleaned_string == reversed_string:
15         print("Yes, your string is a palindrome.")
16     else:
17         print("Sorry, your string is not a palindrome.")
18
19     # Ask if the user wants to continue
20     continue_prompt = input("Do you want to check another string? (yes/no): ").strip().lower()
21     if continue_prompt != 'yes':
22         print("Thank you for using the program!")
23         break # Exit the loop if the user does not want to continue
24
```

Output:

```
C:\Users\hp\PycharmProjects\main\venv\Scripts\python.exe C:\Users\hp\PycharmProjects\main\main2.p
Enter a string to check if it is a palindrome: moM
Yes, your string is a palindrome.
Do you want to check another string? (yes/no): yeS
Enter a string to check if it is a palindrome: DAD
Yes, your string is a palindrome.
Do you want to check another string? (yes/no): yes
Enter a string to check if it is a palindrome: 44
Yes, your string is a palindrome.
Do you want to check another string? (yes/no): Yes
Enter a string to check if it is a palindrome: none
Sorry, your string is not a palindrome.
Do you want to check another string? (yes/no): no
Thank you for using the program!

Process finished with exit code 0
```

5. Write a program which will collect your name, your father's name, your roll number and your subjects (5 Subjects with name and numbers). At the end it will generate a result with your name, your father's name, your details subjects, marks you have obtained with total marks with grade and percentage.

Source Code:

```
1  # Program to collect student information and generate a result
2
3  while True:
4      # Collecting personal details
5      name = input("Enter your name: ")
6      father_name = input("Enter your father's name: ")
7      roll_number = input("Enter your roll number: ")
8
9      total_marks = 0
10     max_marks = 100 # Assuming each subject is out of 100
11     num_subjects = 5 # Total number of subjects
12     subjects = [] # List to hold subject names and marks
13
14     # Collecting subject names and marks
15     for i in range(num_subjects):
16         subject = input(f"Enter the name of subject {i + 1}: ")
17         marks = float(input(f"Enter marks obtained in {subject}: "))
18         subjects.append((subject, marks)) # Store as a tuple (subject, marks)
19         total_marks += marks # Add to total marks
20
21     # Calculating percentage
22     percentage = (total_marks / (num_subjects * max_marks)) * 100
23
24     # Determining grade based on percentage
25     if percentage >= 90:
26         grade = 'A+'
27     elif percentage >= 80:
28         grade = 'A'
29     elif percentage >= 70:
30         grade = 'B'
31     elif percentage >= 60:
32         grade = 'C'
33     elif percentage >= 50:
34         grade = 'D'
35     else:
36         grade = 'F'
```

```

38     # Displaying the result
39     print("\n--- Result ---")
40     print(f"Name: {name}")
41     print(f"Father's Name: {father_name}")
42     print(f"Roll Number: {roll_number}")
43
44     print("\nSubject Details:")
45     for subject, marks in subjects:
46         print(f"{subject}: {marks}")
47
48     print(f"\nTotal Marks Obtained: {total_marks}/{num_subjects * max_marks}")
49     print(f"Percentage: {percentage:.2f}%")
50     print(f"Grade: {grade}")
51
52     # Ask if the user wants to enter another result
53     continue_prompt = input("\nDo you want to enter another result? (yes/no): ").strip().lower()
54     if continue_prompt != 'yes':
55         print("Thank you for using the program!")
56         break # Exit the loop if the user does not want to continue
57

```

Output:

```

C:\Users\hp\PycharmProjects\main\venv\Scripts\python.exe C:\Users\hp\PycharmProjects\main\main2.py
Enter your name: Raazia
Enter your father's name: Imran
Enter your roll number: 009
Enter the name of subject 1: Calculus
Enter marks obtained in Calculus: 90
Enter the name of subject 2: FIT
Enter marks obtained in FIT: 89
Enter the name of subject 3: English
Enter marks obtained in English: 80
Enter the name of subject 4: OHS
Enter marks obtained in OHS: 89
Enter the name of subject 5: PF
Enter marks obtained in PF: 85

--- Result ---
Name: Raazia
Father's Name: Imran
Roll Number: 009

Subject Details:
Calculus: 90.0
FIT: 89.0
English: 80.0
OHS: 89.0
PF: 85.0

Total Marks Obtained: 433.0/500
Percentage: 86.60%
Grade: A

Do you want to enter another result? (yes/no): no
Thank you for using the program!

Process finished with exit code 0

```


5.Generate a table from initial value to final, depending upon the user starting and ending range in matrix form such as:

```
1 2 3 4 5
2 4 6 8 10
3 6 9 12 15
4 8 12 16 20
5 10 15 20
25
```

Source Code:

```
1  # Get user input for starting and ending values
2  start = int(input("Enter the starting value: "))
3  end = int(input("Enter the ending value: "))
4
5  # Generate the multiplication table
6  print("\nMultiplication Table:")
7  for i in range(start, end + 1): # Loop through each row
8      for j in range(1, end + 1): # Loop through each column
9          print(i * j, end="\t") # Print the product, with tab space
10     print() # New line after each row
```

Output:

```
C:\Users\hp\PycharmProjects\main\venv\Scripts\python.exe C:\Users\hp\PycharmProjects\main\main2.py
```

```
Enter the starting value: 1
```

```
Enter the ending value: 5
```

```
Multiplication Table:
```

```
1  2  3  4  5
2  4  6  8  10
3  6  9  12 15
4  8  12 16 20
5  10 15 20 25
```

```
Process finished with exit code 0
```

6. Write a program which will add two square matrices.

Source Code:

```
1  # Define the size of the square matrices
2  n = 2 # You can change this to any size you want
3
4  # Initialize two square matrices with predefined values
5  matrix1 = [[1, 2], [3, 4]] # First matrix
6  matrix2 = [[5, 6], [7, 8]] # Second matrix
7
8  # Initialize a result matrix with zeros
9  result_matrix = [[0 for j in range(n)] for i in range(n)]
10
11 # Add the two matrices
12 for i in range(n): # Loop through rows
13     for j in range(n): # Loop through columns
14         result_matrix[i][j] = matrix1[i][j] + matrix2[i][j] # Add corresponding elements
15
16 # Display the resulting matrix
17 print("Resulting Matrix after Addition:")
18 for i in range(n):
19     for j in range(n):
20         print(result_matrix[i][j], end=' ') # Print each element in the row
21     print() # Move to the next line after each row
22
```

Output:

```
C:\Users\hp\PycharmProjects\main\venv\Scripts\python.exe C:\Users\hp\PycharmProjects\main\main2.py
```

```
Resulting Matrix after Addition:
```

```
6 8
```

```
10 12
```

```
Process finished with exit code 0
```

7. Write a program which will multiply two square matrices.

Source Code:

```
1  # Input size of the square matrices
2  n = int(input("Enter the size of the square matrices: "))
3
4  # Input matrix A
5  print("Enter the elements of matrix A:")
6  A = []
7  for i in range(n):
8      A.append([])
9      for j in range(n):
10         A[i].append(int(input(f"Enter A[{i}][{j}]: ")))
11
12 # Input matrix B
13 print("Enter the elements of matrix B:")
14 B = []
15 for i in range(n):
16     B.append([])
17     for j in range(n):
18         B[i].append(int(input(f"Enter B[{i}][{j}]: ")))
19
20 # Initialize result matrix
21 result = []
22 for i in range(n):
23     result.append([])
24     for j in range(n):
25         result[i].append(0)
26
27 # Matrix multiplication
28 for i in range(n): # Loop through rows of A
29     for j in range(n): # Loop through columns of B
30         for k in range(n): # Loop through rows of B
31             if k < n: # Check index boundary
32                 result[i][j] += A[i][k] * B[k][j]
33
```

```

33
34 # Output the result matrix
35 print("The resulting matrix after multiplication is:")
36 for i in range(n):
37     for j in range(n):
38         if j < n - 1: # Condition to avoid extra space at the end of a row
39             print(result[i][j], end=" ")
40         else:
41             print(result[i][j]) # Print the last element and move to the next line
42

```

Output:

```

C:\Users\hp\PycharmProjects\main\venv\Scripts\python.exe "C:\Users\hp\PycharmProjects\main\main 3.py"
Enter the size of the square matrices: 2
Enter the elements of matrix A:
Enter A[0][0]: 1
Enter A[0][1]: 2
Enter A[1][0]: 3
Enter A[1][1]: 4
Enter the elements of matrix B:
Enter B[0][0]: 4
Enter B[0][1]: 3
Enter B[1][0]: 2
Enter B[1][1]: 1
The resulting matrix after multiplication is:
8 5
20 13

Process finished with exit code 0

```

